

Task 1: Data Summary

Problem or Challenge

This analysis aims to address subgoal 3.4 of goal 3, Health, which revolves around reducing premature mortality from non-communicable diseases through prevention and treatment by one third, as well as promoting mental health and well-being. According to the World Health Organization, the number of diabetic individuals has risen from 200 million in 1990 to 830 million in 2022, with prevalence increasing at a faster rate in low- and middle-income countries. In 2022, diabetes affected 14% of adults aged 18 years and older. Additionally, it is also found that 59% of diabetic adults aged 30 years and older forsake their medication in 2022. Moreover, nearly half of all diabetes-related deaths occur before individuals reach 70 years of age. In 2021, diabetes was responsible for 1.6 million deaths. Diabetes is also known to cause other ailments such as kidney or cardiovascular diseases [1].

Currently, according to the International Diabetes Federation, in 2024, there were approximately 589 million adults living with diabetes, and it is projected to rise to 583 million by 2050 [2].

Target Audience

Adults are the intended audience for this study. Many adults seek strategies to better manage their diabetes, before or in conjunction with professional help. This analysis may help those to attenuate the negative impact of poor diabetes management before they can receive treatment. Additionally, adults also make up most of the world's population [3]; adults are the backbone of society; they are driving the economic growth of the country and raising the next generation; so, it's crucial for them to comprehend the factors behind stable and healthy physical well-being. By analysing the possible factors, adults can make informed, meaningful decisions on the proactive actions that they can take.

Nature of The Dataset

This dataset is derived from the 2015 Behavioral Risk Factor Surveillance System (BRFSS) survey, conducted by the Centers for Disease Control and Prevention (CDC). It contains health-related responses from adults aged 18 and above across the United States. The

scope of this dataset is to understand the correlations and impacts between various factors and diabetes prevalence within this population.

The dataset is structured into four main categories:

Category	Attribute Name	Description	Data type
Demographic	Sex	Gender identity of the individual	Binary, Categorical, Nominal
	Age	Age of the respondent	Categorical, Ordinal
	Education	Highest education level of the respondent	Categorical, Ordinal
	Income	Household income of the respondent	Categorical, Ordinal
Lifestyle	Smoker	Lifetime smoking history	Binary, Categorical, Nominal
	PhysActivity	Leisure physical activity (past 30 days)	Binary, Categorical, Nominal
	Fruits	Daily fruit intake	Binary, Categorical, Nominal
	Veggies	Daily vegetable consumption	Binary, Categorical, Nominal
Health Indicators	HvyAlcoholConsump	Heavy drinking status	Categorical, Ordinal
	Diabetes_binary	Presence of diabetes	Categorical, Ordinal
	HighBP	Presence of high blood pressure	Binary, Categorical, Nominal
	HighChol	Presence of high cholesterol	Binary, Categorical, Nominal
	BMI	Body Mass Index (BMI)	Quantitative, Continuous, Ratio
	Stroke	Presence of stroke	Binary, Categorical, Nominal
	HeartDiseaseorAttack	Presence of coronary heart disease or myocardial infarction	Binary, Categorical, Nominal
	GenHlth	Self-rated general health	Categorical, Ordinal
	MentHlth	Count of poor mental health days (past 30 days)	Quantitative, Discrete, Ratio
	PhysHlth	Days of poor physical health (past 30 days)	Quantitative, Discrete, Ratio
	DiffWalk	Serious difficulty on walking or climbing stairs	Binary, Categorical, Nominal

Healthcare Utilization	CholCheck	Cholesterol screening (past 5 years)	Binary, Nominal	Categorical,
	AnyHealthcare	Health care coverage status	Binary, Nominal	Categorical,
	NoDocbcCost	Inability to afford doctor visit (past 12 months)	Binary, Nominal	Categorical,

The analyses that will be performed on this dataset are descriptive analysis, which focuses on historical analysis; prescriptive analysis, in which courses of action are suggested; and exploratory data analysis (EDA), whereby the focus is on gaining initial insights into the distributions and patterns within the data.

Source of dataset: <https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset/data>

Research Questions:

This analysis aims to uncover the following questions:

1. What role do age and gender play in the risk of diabetes?
2. What is the relationship between diabetes and common medical conditions such as high blood pressure, high cholesterol, stroke, heart disease, and difficulty walking?
3. How do smoking habit and physical activity influence the likelihood of diabetes?

(Word count: 457 words)

Task 2: Data Memo

2.1 What role do age and gender play in risk of diabetes?

2.1.1 Analysis

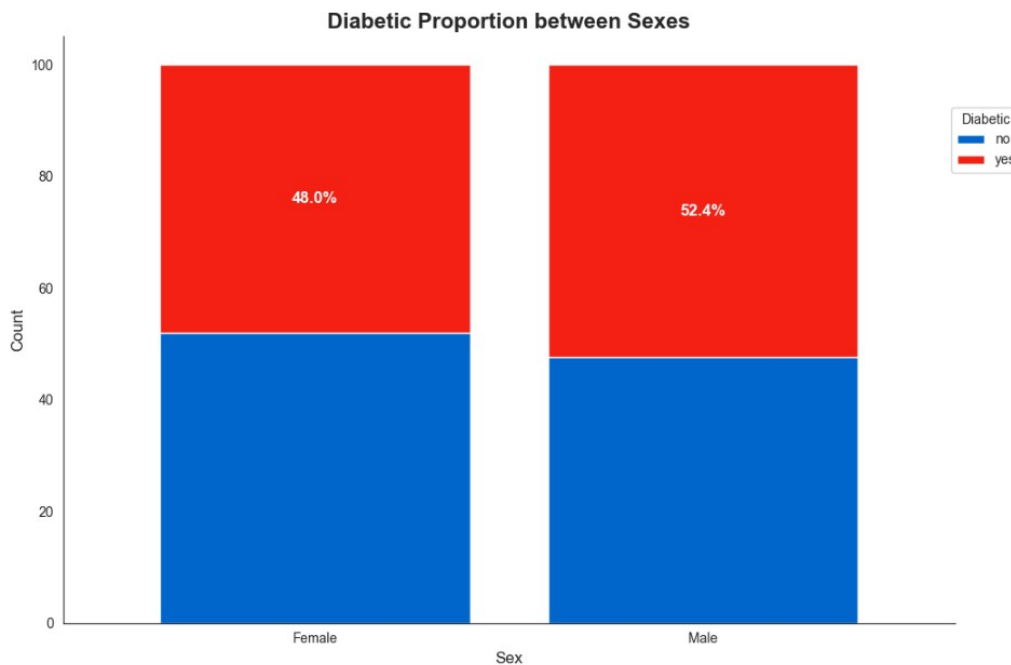


Figure 2.1.1.1: 100% stacked bar chart on diabetic proportion between sexes

It is found that, relative to their group size, men are more likely to have diabetes compared to their female counterparts. According to [4], men tend to develop type 2 diabetes at a lower body weight than women. Additionally, physiologically, men tend to store fat in their bellies, which is a known risk factor. On the other hand, the risk of diabetes is lower amongst women, but they do have a higher risk of complications—heart disease, vision loss, kidney disease, and depression—compared to their male counterparts. Interestingly, women who are from African American, Hispanic/Latina, American Indian/Alaska Native, and Asian/Pacific Islander backgrounds exhibit a higher prevalence of diabetes compared to their White counterparts.

Another interesting variable to consider is age groups; by including age groups, we may be able to uncover unique patterns between genders across various age groups.

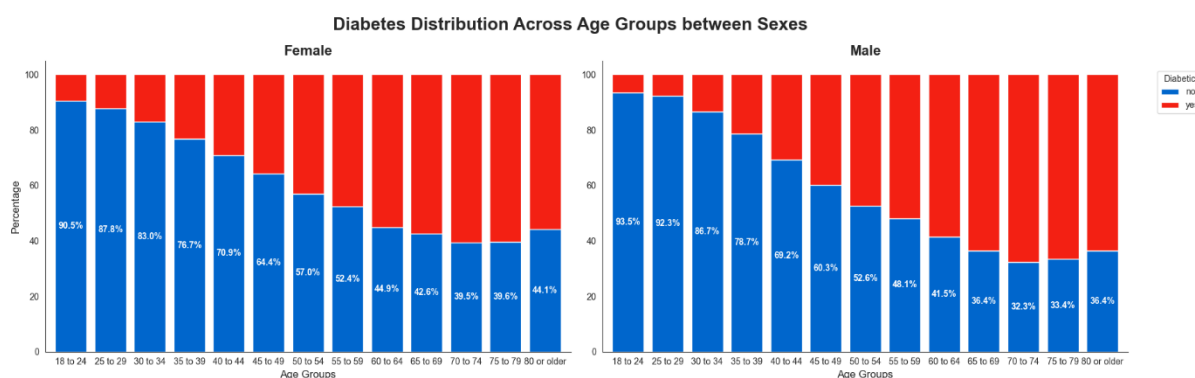


Figure 2.1.1.2: 100% stacked bar chart on diabetes distribution across age groups between sexes

In both genders, the proportion of non-diabetic individuals declines as age increases, but a sharper increase is found amongst males. According to [5], the risk of type 2 diabetes goes up with age, especially after age 35, which is observable in the sample. According to [6], seniors can be more susceptible to inactivity, sleep deprivation, and metabolism-related issues. Inactivity is prevalent amongst seniors [7], with only 48% remaining active between the ages of 75 and 84, and just 26% at age 85 and above. Such inactivity leads to underutilisation of glucose in the body, leading to excessive glucose in the blood [6]. Additionally, poor sleep quality is more frequent amongst older adults due to age-related factors such as multiple chronic health conditions, the use of several medications, psychological or social issues, and specific sleep disorders [8]. According to [8], obstructive sleep apnea (OSA) is estimated to affect 70% of older males and 56% of older females, which may explain the observable pattern. A link between obstructive sleep apnea (OSA) and diabetes has been observed, whereby individuals with OSA have a higher prevalence of diabetes mellitus. According to [9], sleep deprivation and poor sleep quality are linked to blood sugar disturbances and increased insulin resistance, both of which lead to greater fluctuations of glucose levels.

Interestingly, there is a slight increase in non-diabetic individuals amongst those aged 80 and above. Since most research supports an increased risk of diabetes with age, this anomaly may simply reflect a coincidental distribution within the sample. Additionally, we can see that in younger age groups, the proportion of diabetics is higher in females than in males. Young women aren't universally more likely to get diabetes, but certain factors do affect their chances. Oestrogen protects premenopausal women from type 2 diabetes by increasing insulin sensitivity and glucose-stimulated insulin secretion. So, women who experience menopause prematurely are associated with an increased risk of type 2 diabetes [10].

2.1.2 Recommendation

Adults, especially men, should practise a balanced diet as they are at a greater risk of developing diabetes due to their physiological tendency to store fat in the abdomen. Individuals can follow the "Quarter-Quarter-Half" method, where half of the plate is dedicated to fruits and vegetables, while the remaining quarters are reserved for protein and carbohydrates. This approach ensures healthier, balanced eating habits at each meal.

The government should advocate for annual health check-ups, especially for adults over 40, as the proportion of diabetics starts to pick up in both genders. Generally, health screenings include blood glucose tests, blood pressure checks, and BMI monitoring, providing individuals with valuable information about their current health. This information helps them make informed decisions about their diet, lifestyle, and physical activity levels, encouraging them to take proactive steps in preventing or managing diabetes. Early detection through regular screenings can also reduce the risk of complications by enabling timely medical intervention. To increase accessibility, governments can collaborate with local clinics and community centres to offer subsidised or mobile screening services, especially in rural areas where healthcare access may be limited.

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2.2 What is the relationship between diabetes and common medical conditions such as high blood pressure, high cholesterol, stroke, heart disease, and difficulty walking?

2.2.1 Analysis

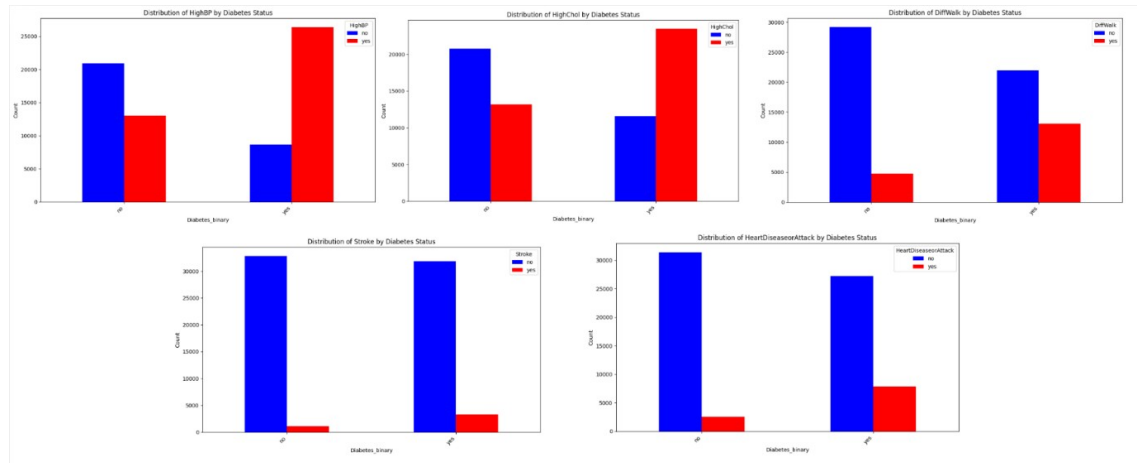


Figure 2.2.2.1: Grouped Bar Chart - Comparison of Medical Conditions Between Diabetics and non-Diabetics

Based on the charts above, it can be seen that for individuals with diabetes, a distinct majority also have **High Blood Pressure** and **High Cholesterol**. In stark contrast, for the non-diabetic population, a large majority do not have these conditions. This demonstrates a strong and visually apparent overlap between them.

For the other conditions which includes **Heart Disease or Attack**, **Stroke**, and **Difficulty Walking**, the overall prevalence is much lower in comparison. Within the diabetic group, most individuals do not have these conditions. However, the critical insight comes from comparing the diabetic and non-diabetic groups. For each of these three conditions, the number of affected individuals is visibly higher among those with diabetes than those without, suggesting a notable, though less frequent, co-occurrence.

Medical Conditions	HighBP	HighChol	HeartDisease orAttack	Stroke	Difficulty Walking
Chi-squared	9557.3230	5467.1781	2964.4036	1039.1216	4924.7961
P-value	0.0	0.0	0.0	0.0	0.0
Degree of Freedom	1	1	1	1	1
Cramer's V	0.3720	0.2813	0.2072	0.1226	0.2670
Rank of Association	1	2	4	5	3

Table 2.2.2.1: Chi-Squared and Cramer's V Statistics for Associations Between Diabetes and Selected Medical Conditions

To statistically validate and quantify the relationships observed in the charts, a Chi-squared test of independence was performed for each condition against diabetes status. The results, summarized in the table above, provide quantitative proof of the associations.

The P-value for every condition is approximately 0.0 (more formally, $p < 0.001$), indicating that the observed relationships are extremely statistically significant. This allows us to confidently conclude that the associations are not a result of random chance.

Furthermore, Cramer's V is used to measure the strength of these associations, allowing for a standardized comparison. This reveals that High Blood Pressure has the strongest association with diabetes, followed by High Cholesterol. The "Rank of Association" in the table summarizes this, identifying the most and least impactful comorbidities in terms of their statistical link to diabetes within this dataset.

On top of that, the correlation heatmap below provides a deeper insight into the relationships between these conditions themselves. This helps to illustrate that the pathway to severe outcomes like a heart attack or stroke is not a direct effect of diabetes alone, but rather a result of a cluster of interrelated risk factors especially high blood pressure and cholesterol.

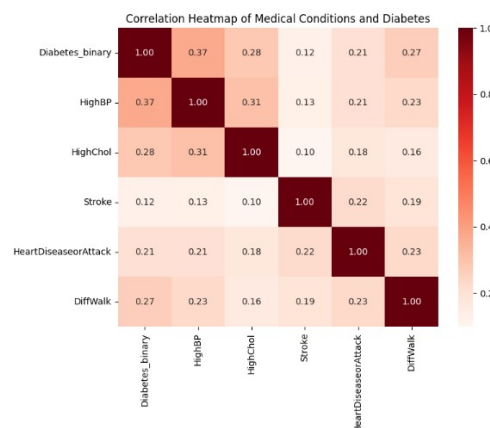


Figure 2.2.2.2: Correlation Heatmap Between Diabetes and Selected Medical Conditions

To explain the associations found in the data, a review of established medical literature was conducted. First of all, the strong link between diabetes, high blood pressure, and high cholesterol is a well-documented phenomenon. These three conditions often cluster together in what is known as metabolic syndrome [11]. When present together, they create a synergistic effect, meaning they amplify each other's harmful impact and dramatically increase the risk of cardiovascular disease compared to having any one condition alone.

The progression can be understood by starting with high blood sugar, or hyperglycemia, caused by diabetes [12]. Over time, excess glucose damages the delicate inner lining of blood vessels. This damage not only makes the vessels more susceptible to the strain of high blood pressure but also allows harmful cholesterol to build up within the vessel walls [13]. Together, these processes accelerate atherosclerosis, also known as the hardening and narrowing of arteries due to plaque buildup [14].

Atherosclerosis is the direct underlying cause of many serious cardiovascular events. When plaque obstructs the arteries supplying blood to the heart, a heart attack may occur. Similarly, when it blocks an artery leading to the brain, it results in an ischemic stroke [14]. This explains why the data shows a significantly higher number of heart attacks and strokes among individuals with diabetes.

Lastly, the strong association with difficulty walking often reflects a later stage of this disease progression. This disabling symptom is typically the result of either nerve damage and poor circulation in the legs known as peripheral artery disease, commonly caused by long-term diabetes or the lasting physical impairments following a stroke [15].

2.2.2 Recommendation

Since diabetes, high blood pressure, and high cholesterol are strongly linked and often appear together, society needs to take a more joined-up approach in tackling them. In healthcare, this means that if a person is diagnosed with one of these conditions, they should automatically be checked for the others. Early detection allows doctors to manage the overall risk instead of treating each problem separately. Beyond clinics, the wider community also has a role to play. Public health campaigns should move away from warning people about single diseases in isolation and instead highlight the dangers of having this “trio” of conditions together, which can lead to serious outcomes like heart disease, stroke, or even difficulty walking later in life. By framing the message around combined risk, the public is more likely to understand the urgency and take preventive steps.

(Word count: 725 words)

2.3 How do smoking habit and physical activity influence the likelihood of diabetes?

2.3.1 Analysis

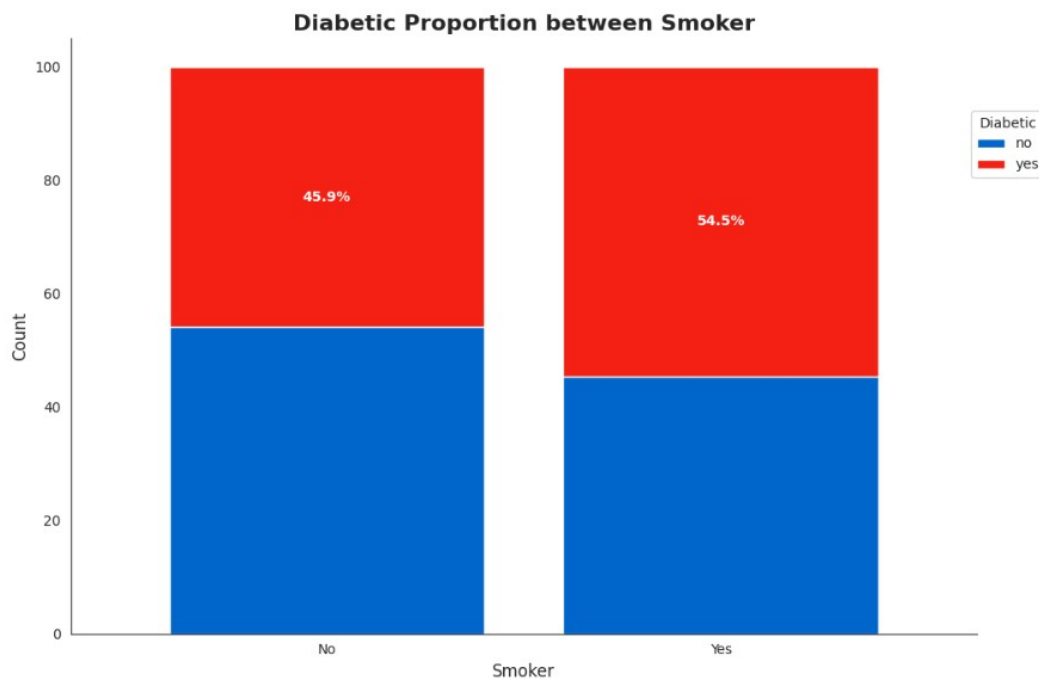


Figure 2.3.1.1: 100% stacked bar chart on diabetic proportion between smoker

Based on the stacked bar chart, individuals who have smoked at least 100 cigarettes in their lifetime show a higher rate of diabetes compared to those who have smoked fewer than 100 cigarettes. This notable difference suggests that smoking is an important risk factor for developing diabetes. According to [16], smokers are 30% to 40% more likely to develop type 2 diabetes than non-smokers. The pancreas produces insulin to regulate blood sugar levels by allowing glucose to be used as fuel for body cells; however, nicotine interferes with this process and causes insulin resistance [17]. If the cells do respond to insulin effectively, glucose accumulates in the blood, resulting in elevated blood sugar levels. Simultaneously, nicotine triggers inflammation in the body, further disrupting the regulation of glucose [16]. These mechanisms may account for the higher prevalence of diabetes observed amongst smokers in the dataset.

This not only fits with the dataset's pattern but also emphasizes how crucial smoking cessation programs are to lowering the incidence of diabetes.

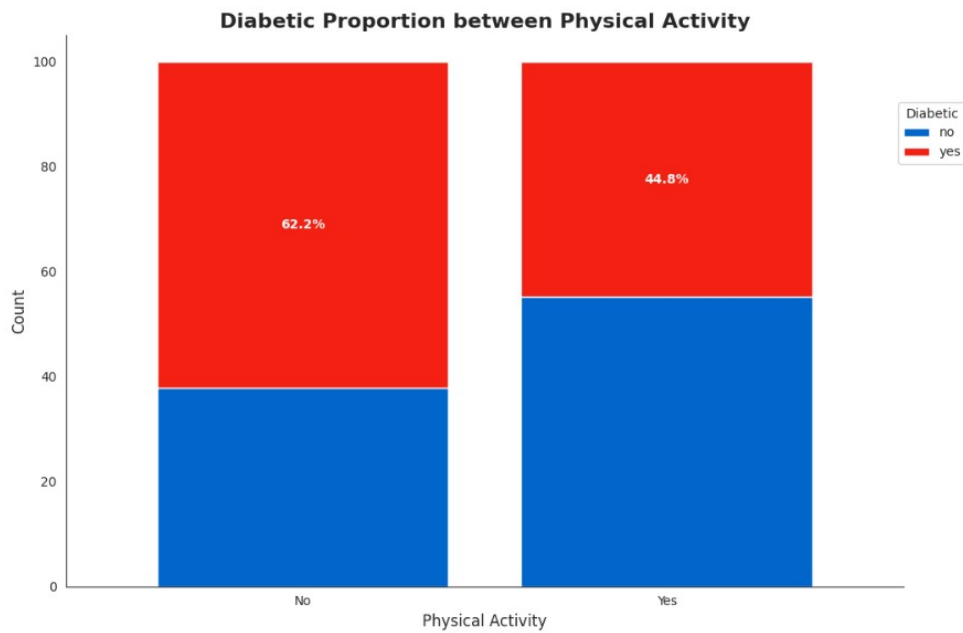


Figure 2.3.1.2: 100% stacked bar chart on diabetic proportion between physical activity

A higher proportion of diabetes is observed among individuals who do not engage in physical activity outside of work, compared to those who physically active. According to [18], individuals who participate in physical activities have approximately 30% to 40% lower risk of developing diabetes compared to sedentary individuals; physical activities lower blood sugar levels by decreasing insulin resistance and making the body more sensitive to insulin [19]. Thus, cells such as muscle cells can absorb glucose from the bloodstream more efficiently. To summarise, physical activity improves the metabolic function and reduces the risk of developing diabetes.

Moreover, physical activity plays a crucial role in maintaining a healthy body weight. As found in [20], abdominal fat is one of the major factors that causes insulin resistance. Regular exercise increases the body's capacity for managing blood sugar by reducing and preventing the accumulation of excess abdominal fat. Hence, physical activity not only lowers the risk of diabetes but also supports overall metabolic health.

2.3.2 Recommendation

There is no safe number of cigarettes, so smokers are strongly encouraged to quit. However, nicotine addiction is difficult to overcome; Individuals can gently ease into nicotine-free by slowly reducing the number of cigarettes per day. To support smoking cessation, nicotine replacement products such as gum, patches, and lozenges may be helpful. In addition, hospitals or the community can encourage smokers attempting cessation to participate

counselling sessions or support groups; they can receive guidance from professionals and motivation through mutual encouragement, helping them to stay committed throughout the process.

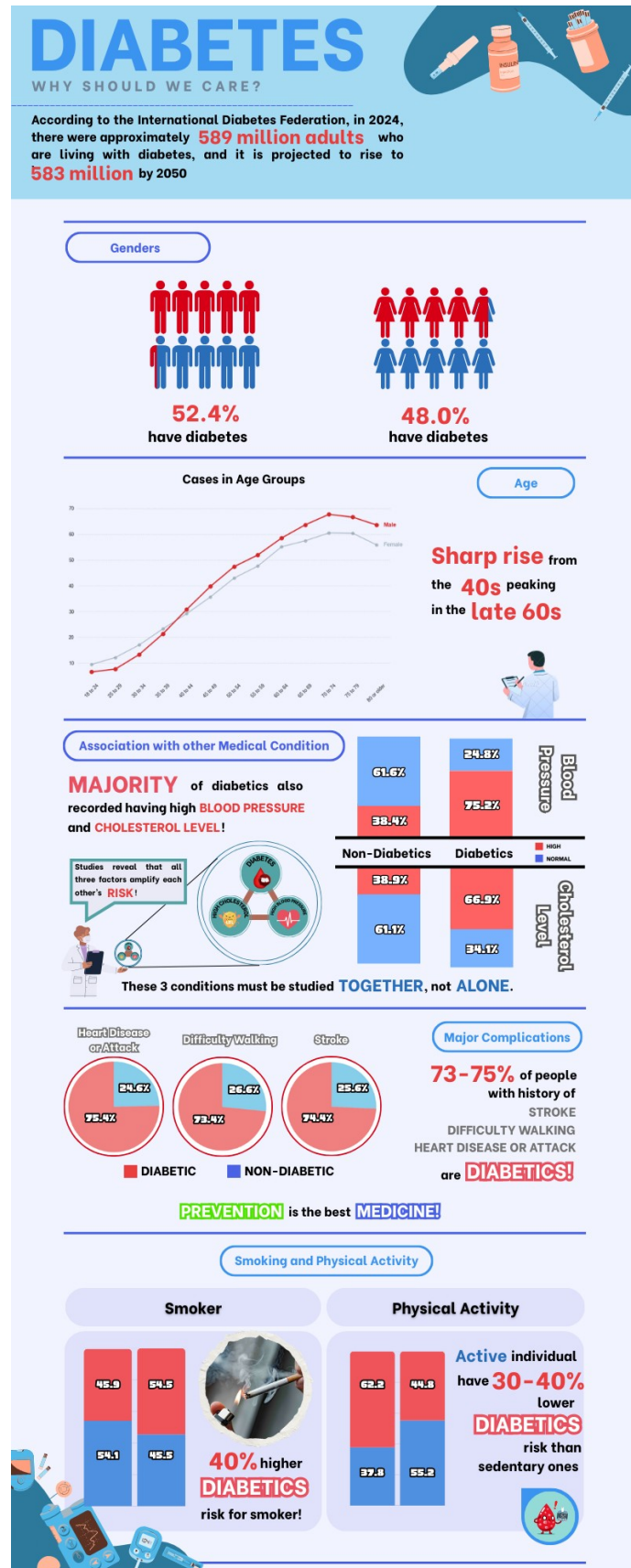
Additionally, according to the Centers for Disease Control and Prevention (CDC), individuals are encouraged to engage in at least 150 minutes of moderate-intensity physical activity, such as jogging, brisk walking, or dancing per week. To maintain discipline, it is important to start with realistic goals and gradually increase the intensity instead of starting with extreme exercises. Even standing up or walking for a few minutes after long periods of sedentary behaviour can help reduce overall sedentary time.

To combat widespread inactivity, local governments can host community exercise programs for the public to participate. The shared experience cultivates a sense of belonging and mutual encouragement, which can significantly increase participation compared to solitary workouts. The presence of others fosters accountability, companionship, and a collective momentum that sustains long-term engagement. Governments should also invest more in building and maintaining public gymnasiums, outdoor gyms, and exercising equipment in park, improving accessibility and subsequently, promotes physical activity in the community.

(Word count: 573 words)

(Total word count: 1943 words)

Task 3: Infographic and Presentation



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